STUDY EXAMPLES

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Problem 1. There are $n \geq 2$ segments in the plane such that every two segments cross, and no three segments meet at a point. Geoff has to choose an endpoint of each segment and place a frog on it, facing the other endpoint. Then he will clap his hands $n-1$ times. Everytime he claps, each frog will immediately jump forward to the next intersection point on its segment. Frogs never change the direction of their jumps. Geoff wishs to place frogs in such a way that no two of them will ever ocupy the same intersection point at the same time.

(1) Prove that Geoff can always fullfil his wish when n is odd.
(2) Prove that Geoff can never fullfil his wish when n is even.

Problem 2. Find all positive integers n for which each cell of the $n \times n$ table can be filled with one of the letters I, M and O such that:

(1) in each row and each column, one third of the entries are I, one third are M and one third are O;
(2) in any diagonal, if the number of entries on the diagonal is a multiple of 3, then one third of the entries are I, one third are M and one third are O.

Problem 3. A positive integer is written in each cell of an 8×8 board. For any tiling of the board with dominoes, the sum of numbers on each domino is different. Can it happen that the largest number on the board is at most 32?

Problem 4. Prove that there are infinitely many positive integers n such that it is possible to place 1 to $3n$ into n groups (each group has 3 numbers) and the sum of the two smaller numbers in each group equals the third.

Problem 5. There is a positive number n on the blackboard. Each time we can select a number $a > 1$ on the blackboard, erase it, and write down all of the positive divisors of a (except for the number a itself). Duplicates are allowed. After a while, there are n^2 numbers on the blackboard. Find all possible values of n.

Problem 6. The cells of a 8×8 table are initially white. Alice and Bob play a game. First Alice paints n of the fields in red. Then Bob chooses 4 rows and 4 columns from the table and paints all fields in them in black. Alice wins if there is at least one red field left. Find the least value of n such that Alice can win the game no matter how Bob plays.

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Problem 7. A cake has the form of an $n \times n$ square composed of n^2 unit squares. Strawberries lie on some of the unit squares so that each row or column contains exactly one strawberry; call this arrangement A. Let B be another such arrangement. Suppose that every grid rectangle with one vertex at the top left corner of the cake contains no fewer strawberries of arrangement B than of arrangement A. Prove that arrangement B can be obtained from A by performing a number of switches, defined as follows: A switch consists in selecting a grid rectangle with only two strawberries, situated at its top right corner and bottom left corner, and moving these two strawberries to the other two corners of that rectangle.